Statistics and Data Analysis for Engineers

A Practical, Python-Oriented Introduction



Richard Brooks

VIA University College

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Part I

Data and Probability Models

Chapter 1 Data and Descriptive Statistics

1.1 Engineering question and learning outcomes

How can measurements be summarised without hiding important variation?

1.2 Populations, samples, and data types

1.3 Numerical summaries

1.4 Visualising distributions and relationships

1.5 Worked engineering example

1.6 Python in practice

1.7 Common pitfalls and chapter summary

1.8 Exercises

Chapter 2 Probability for Data Analysis

2.1 Engineering question and learning outcomes

How can uncertainty in an engineering decision be represented and updated?

2.2 Sample spaces, events, and probability rules

2.3 Conditional probability and independence

2.4 Total probability and Bayes' rule

2.5 Worked engineering example

2.6 Python in practice

2.7 Common pitfalls and chapter summary

2.8 Exercises

Chapter 3 Random Variables and Discrete Distributions

3.1 Engineering question and learning outcomes

How can counts and binary outcomes be modelled probabilistically?

3.2 Random variables, PMFs, and CDFs

3.3 Expectation and variance

3.4 Bernoulli and binomial models

3.5 Poisson models

3.6 Python in practice

3.7 Common pitfalls and chapter summary

3.8 Exercises

Chapter 4 Continuous Distributions in Practice

4.1 Engineering question and learning outcomes

How can continuous measurements, waiting times, and lifetimes be modelled?

4.2 Densities, distribution functions, and quantiles

4.3 Uniform, normal, and exponential models

4.4 Model checking and QQ plots

4.5 Worked engineering example

4.6 Python in practice

4.7 Common pitfalls and chapter summary

4.8 Exercises

Part II

Sampling and Statistical Inference

Chapter 5 Sampling Distributions and the Central Limit Theorem

5.1 Engineering question and learning outcomes

Why do estimates vary from sample to sample, and what controls that variation?

5.2 Statistics as random variables

5.3 The sampling distribution of the mean

5.4 Standard error

5.5 The central limit theorem

5.6 Python simulation

5.7 Common pitfalls and chapter summary

5.8 Exercises

Chapter 6 Estimation and Confidence Intervals

6.1 Engineering question and learning outcomes

How precisely has a population quantity been estimated from a sample?

6.2 Estimators, estimates, bias, and precision

6.3 Intervals for a mean

6.4 Intervals for a proportion

6.5 Coverage and interpretation

6.6 Python in practice

6.7 Common pitfalls and chapter summary

6.8 Exercises

Chapter 7 Hypothesis Testing I: Principles and One-Sample Tests

7.1 Engineering question and learning outcomes

Does a process provide evidence against a specified target or baseline?

7.2 Hypotheses, significance levels, and p-values

7.3 Type I and type II errors

7.4 One-sample tests for a mean and proportion

7.5 Effect size and practical importance

7.6 Python in practice

7.7 Common pitfalls and chapter summary

7.8 Exercises

Chapter 8 Hypothesis Testing II: Comparing Two Groups

8.1 Engineering question and learning outcomes

Do two processes, systems, or treatments differ in a practically meaningful way?

8.2 Independent and paired designs

8.3 Welch's two-sample t-test

8.4 The paired t-test

8.5 Comparing two proportions

8.6 Python in practice

8.7 Common pitfalls and chapter summary

8.8 Exercises

Chapter 9 Analysis of Variance

9.1 Engineering question and learning outcomes

How can several group means be compared without inflating the false-positive rate?

9.2 The one-way ANOVA model

9.3 Between-group and within-group variation

9.4 The F-test and its assumptions

9.5 Post-hoc comparisons

9.6 Python in practice

9.7 Common pitfalls and chapter summary

9.8 Exercises

Part III

Modelling and Integrated Analysis

Chapter 10 Simple Linear Regression

10.1 Engineering question and learning outcomes

How can a linear relationship between two quantitative variables be modelled and interpreted?

10.2 The regression model and least squares

10.3 Correlation and coefficient of determination

10.4 Inference for the slope

10.5 Residual diagnostics and prediction

10.6 Python in practice

10.7 Common pitfalls and chapter summary

10.8 Exercises

Chapter 11 Categorical Data and Chi-Square Tests

11.1 Engineering question and learning outcomes

How can observed category counts be compared with a model or across groups?

11.2 Counts, proportions, and expected counts

11.3 Chi-square goodness-of-fit tests

11.4 Chi-square tests of independence

11.5 Residuals and follow-up interpretation

11.6 Python in practice

11.7 Common pitfalls and chapter summary

11.8 Exercises

Chapter 12 Integrated Statistical Analysis and Exam Preparation

12.1 Engineering question and learning outcomes

How can a defensible analysis be carried from raw data to an engineering recommendation?

12.2 From question to data audit

12.3 Method selection

12.4 Assumptions, uncertainty, and limitations

12.5 Communicating a recommendation

12.6 Integrated Python case study

12.7 Common pitfalls and chapter summary

12.8 Exercises

References

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Appendix A: Python and Data Setup

This appendix supports the preliminary STA1 setup session. It will cover the course environment, reproducible random-number generation, data paths, DataFrame inspection, plotting conventions, and notebook reporting workflow.

A.1 Recommended environment

A.2 Core packages

A.3 Reproducible analysis

A.4 Loading course data

A.5 Notebook and reporting workflow